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Use of Higher Order Schemes for Flow Computations

An internship report submitted
in partial fulfillment for the award of the degree of

Bachelor of Technology

in

Aerospace Engineering

by

Ajay Daniel Pravin



**Department of Aerospace Engineering
Indian Institute of Space Science and Technology
Thiruvananthapuram, India**

June-Aug 2022

Certificate

This is to certify that the internship report titled *Use of Higher Order Schemes for Flow Computations* submitted by **Ajay Daniel Pravin**, to the Indian Institute of Space Science and Technology, Thiruvananthapuram, in partial fulfillment for the award of the degree of **Bachelor of Technology** is a bona fide record of the original work carried out by him/her under my supervision. The contents of this internship report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Dr. Manoj T Nair

Professor

Department of Aerospace Engineering

Indian Institute of Space Science and Tech-
nology

Thiruvananthapuram

Place: Thiruvananthapuram

Date: June-Aug 2022

Declaration

I declare that this internship report titled *Use of Higher Order Schemes for Flow Computations* submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Aerospace Engineering** is a record of the original work carried out by me under the supervision of **Dr. Manoj T Nair**, and has not formed the basis for the award of any degree, diploma, associateship, fellowship, or other titles in this or any other Institution or University of higher learning. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

Place: Thiruvananthapuram

Date: June-Aug 2022

Ajay Daniel Pravin

(SC19B042)

This internship report is dedicated to my parents who instilled a sense of explorative curiosity, and all my teachers since who armed me with the tools to apply it. . .

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Firstly, I would like to thank my advisor and guide, Dr. Manoj T Nair, for his constant support and encouragement throughout the internship. He provided useful tools and study materials necessary for understanding the subject and completing the project.

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Lastly, I would like to thank my parents who have always been supportive of my endeavours.

Ajay Daniel Pravin

Abstract

In this project, different types of discretization schemes for Finite Difference using Taylor Series expansions are described and derived. Error analysis is conducted on the derived schemes and determines their accuracy. To do this, Fourier Error Analysis is utilized. After conducting this analysis on the first derivative of the function of interest, a transfer function equating the modified wavenumber after analysis and the wavenumber of the actual solution is obtained. After plotting and comparing the relations between these wave numbers for different schemes, the relative accuracy between schemes can be understood. Furthermore, the deviation curves between schemes of same order can be plotted which are used for further deciding the advantages and disadvantages of each scheme.

Contents

List of Figures	xiii
Nomenclature	xv
1 Introduction	1
2 Literature Review	2
2.1 Significance of Order in a scheme	2
2.2 Higher order scheme usage at boundary nodes	2
2.3 Compact schemes	5
2.4 Analysis of Errors	5
2.5 Spectral Methods	6
3 Formulation of Schemes	8
4 Graphical Comparisons between Schemes	13
5 Results and Conclusions	15
5.1 Scheme deviations	15
Bibliography	17
Appendices	19
A MATLAB Code	19
A.1 For CDS	19
A.2 For Tridiagonal schemes	20
A.3 For Pentadiagonal schemes and 4th order Spectral	21

List of Figures

2.1	Couette's Flow	3
2.2	Node positions	3
4.1	CDS w_m vs w	13
4.2	Tridiagonal schemes w_m vs w	13
4.3	Pentadiagonal schemes w_m vs w	14
4.4	Complete Comparisons with Spectral w_m vs w	14
5.1	Deviation 4^{th} order: Tridiagonal vs CDS	15
5.2	Deviation 4^{th} order: Spectral vs CDS	16
5.3	Deviation 4^{th} order: Spectral vs Tridiagonal	16
5.4	Deviation 6^{th} order: Tridiagonal vs CDS	16
5.5	Deviation 8^{th} order: Pentadiagonal vs Tridiagonal	17
5.6	Deviation: 4^{th} Spectral vs 10^{th} Pentadiagonal	17

Nomenclature

<i>FDS</i>	Forward Difference Scheme
<i>BDS</i>	Backward Difference Scheme
<i>CDS</i>	Central Difference Scheme
<i>1D</i>	One dimensional
<i>2D</i>	Two dimensional
<i>3D</i>	Three dimensional
<i>CFD</i>	Computational Fluid Dynamics
<i>FT</i>	Fourier Transform

Chapter 1

Introduction

For this project and presentation, different schemes of varying complexities and orders will be derived in order to compare their accuracy. Another aim of the project will be to show that formal accuracy is not the only basis for determining the complete accuracy of the scheme.

To prove the above, we will need to introduce the Fourier Error Analysis which will give us discretization errors. The relations between the discretization errors and Truncation errors will shed light on the difference between the actual solution and the discretized solution for the derivative at a node.

The assumptions we make throughout the project are noted below:

1. All the approximations are conducted on a uniform grid. The grid size is denoted by either ' h ' or ' Δx ' in the report.
2. All the functions are assumed to be periodic. This is to prevent any issues with the boundary since we want to analyze CDS over FDS and BDS.
3. All the formulations and calculations are done only on the first derivative of the function.

Chapter 2

Literature Review

2.1 Significance of Order in a scheme

The order of a scheme will indicate the rate at which the error will decrease when the grid /step size (Δx) is reduced. Therefore the rate of convergence will be faster for a scheme with higher accuracy.

General truncation error form is given as:

$$\epsilon_\tau = (\Delta x)^m \alpha_{m+1} + (\Delta x)^{m+1} \alpha_{m+2} + (\Delta x)^{m+2} \alpha_{m+3} + \dots + (\Delta x)^n \alpha_{n+1}$$

where α denotes higher order derivatives multiplied by some constant value

Note that the order of the scheme says nothing about error when the grid size (Δx) is increasing (coarser grid). Using higher order schemes at larger step sizes can give more inaccurate solutions when compared to lower ordered schemes.

Similarly for turbulent flow simulation, due to the variation of solution of a range of scales, the Taylor series approximation may be misleading and higher order solution error may not be significantly smaller than a lower order scheme.

2.2 Higher order scheme usage at boundary nodes

Example is taken for Couette's flow with velocity u and gradient $\frac{\partial u}{\partial y}$

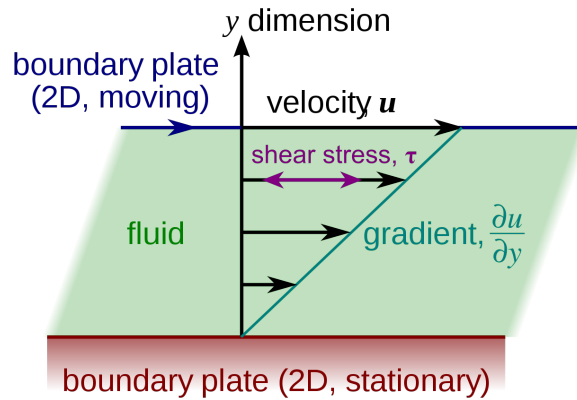


Figure 2.1: Couette's Flow

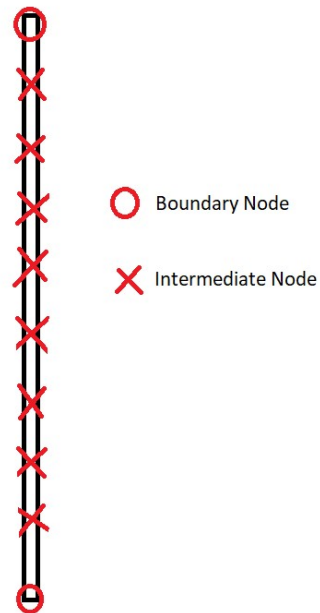


Figure 2.2: Node positions

To compute the derivative value at intermediate points, we can use classic CDS. We can only use a biased scheme at boundary nodes as points beyond the boundary do not exist/are undefined.

Classical FDS/BDS can be used at these boundary points. But how do we match the formal accuracy (order of truncation error) of the biased scheme to be the same as that of the central schemes used at the intermediate points?

Central difference scheme at node x_i

$$\left(\frac{\partial u}{\partial y}\right)_i = \frac{u_{i+1} - u_{i-1}}{2\Delta y}$$

This is of order two and uses two nodes y_{i-1} and y_{i+1} for computation.

However, when using a two-point FDS (to get the boundary node derivative value), we end up with the following scheme

$$\left(\frac{\partial u}{\partial y}\right)_i = \frac{u_{i+1} - u_i}{\Delta y}$$

This is of order one which is lower than the above central scheme for intermediate points.

To solve this problem, we take more node points for our computations. The easiest way to determine the number of nodes for a given scheme with a required order of accuracy is mentioned below.

Let 'n' be the number of nodes, 'p' be the order of the derivative to be computed 'q' be the required order of accuracy.

1. Biased schemes (Upwind/Downwind)

$$n = p + q$$

2. Central schemes

$$n = p + q - 1$$

From this, to match our central difference scheme (order 2) for finding $\frac{\partial u}{\partial y}$, we use p=1 and q=2 giving us a total of 3 points needed for our upwind/downwind schemes

At the boundary node 'i', the other points of interest will be therefore i+1 and i+2, giving a total of 3 points for computation.

By equating the coefficients at LHS and RHS, we can obtain the values for a,b and c. The final scheme is obtained after solving the solution matrix.

Using this general finite difference method, we can formulate the derivative of any order

for any required level of accuracy.

2.3 Compact schemes

Here we use a combination of variable values at the nodes as well as the derivative values at some points. Pade schemes are used for formulation

For up to order of six, a general compact centered approximation is given by:

$$\alpha \left(\frac{\partial \phi}{\partial x} \right)_{i+1} + \beta \left(\frac{\partial \phi}{\partial x} \right)_i + \alpha \left(\frac{\partial \phi}{\partial x} \right)_{i-1} = \beta \frac{\phi_{i+1} - \phi_{i-1}}{2\Delta x} + \gamma \frac{\phi_{i+2} - \phi_{i-2}}{4\Delta x}$$

When comparing classical central difference schemes and Pade schemes of the same order, we can see that the latter uses fewer computational nodes (therefore it is a compact scheme).

2.4 Analysis of Errors

The discretized equation we obtain after finite difference is merely an approximation as we ignore the truncation error. Therefore the exact solution of the differential at that point (ϕ) will not satisfy the discretized form.

$$\mathcal{L}(\phi) = \mathcal{L}_h(\phi) + \tau_h$$

where $\mathcal{L}$ represents the differential equation, $\mathcal{L}_h$ represents the discretized system and τ_h represents the truncation error of a grid 'h'.

Now we introduce a discretization error ϵ_h^d , to equate the exact solution ϕ and the discretized exact solution ϕ_h

$$\phi = \phi_h + \epsilon_h^d$$

$$\mathcal{L}_h(\epsilon_h^d) = -\tau_h$$

From the discretization error formulation, we can understand that the rate of convergence matters due to the 'refining' of the grid 'h' (i.e. the grid size becomes finer)

2.5 Spectral Methods

Using spectral methods, we can interpolate a function $f(x)$ which is defined on a continuous variable x , by replacing the variable x_i . Initially the function is depicted as a discrete Fourier series.

$$f(x_i) = \sum_{q=-N/2}^{N/2-1} \hat{f}(k_q) e^{ik_q x}$$

We can obtain the Fourier coefficients by inverting the above equation.

$$\hat{f}(k_q) = \frac{1}{N} \sum_{i=1}^N f(x_i) e^{-ik_q x_i}$$

After interpolating using the starting equation for $f(x_i)$ and substituting the Fourier coefficient, we can obtain the value of the differential. Note that $k_q = 2\pi q / \Delta x N$ where N is the number of grid points.

$$\frac{df}{dx} = \sum_{q=-N/2}^{N/2-1} ik_q \hat{f}(k_q) e^{ik_q x}$$

We can see that the discrete variable (x_i) is replaced by x which means that the function is defined not only for the grid points, but for every single value of x .

Spectral methods are suited to simulate turbulent flow because they provide exponentially less error when compared to finite difference error for large number of N . Note that the function is also assumed periodic for this particular derivation.

Discretization errors for Spectral Methods

If we use some discretization scheme on the above function, there will be some deviation w.r.t the actual value. This is because the finite difference methods used are equivalent to changing the actual wavenumber (k) by a 'modified' wavenumber (k_m).

This modified wavenumber can be obtained for any scheme on doing the Fourier Analysis and performing the Fourier Transform on the differential. The e^{ik} terms can be split into their respective sines and cosines using Euler's Law for Complex numbers ($e^{ix} = \cos(x) + i\sin(x)$) effectively making it a transfer function.

Chapter 3

Formulation of Schemes

Taking f as the function value at the node x_i . We want to calculate the first derivative $\frac{\partial f}{\partial x}$. We express the schemes in terms of a generalized Pade scheme.

$$\beta\left(\frac{\partial f}{\partial x}\right)_{i+2} + \alpha\left(\frac{\partial f}{\partial x}\right)_{i+1} + \frac{\partial f}{\partial x_i} + \alpha\left(\frac{\partial f}{\partial x}\right)_{i-1} + \beta\left(\frac{\partial f}{\partial x}\right)_{i-2} = c\frac{f_{i+3}-f_{i-3}}{6\Delta x} + b\frac{f_{i+2}-f_{i-2}}{4\Delta x} + a\frac{f_{i+1}-f_{i-1}}{2\Delta x}$$

Now applying the Fourier analysis, we can obtain the resultant wavenumber in terms of e^{ik} terms. After expanding and simplifying the resultant wavenumber, we get the numerator and denominator in terms of sines and cosines. This resultant wavelength resembles a transfer function for each scheme which we in-turn compare with the actual differentiation of the function.

$$\frac{df}{dx} = \sum_{q=-N/2}^{N/2-1} ik_q \hat{f}(k_q) e^{ik_q x}$$

From the derivative, we can get the following modified wavenumber expression

$$w_m = \frac{a \sin(w) + (b/2) \sin(2w) + (c/3) \sin(3w)}{1 + 2\alpha \cos(w) + 2\beta \cos(2w)}$$

Now we can compare the accuracy of the different types of schemes with a range of orders. Then finally we can observe the results by comparing the deviation trends.

3.0.1 Central Difference Schemes

2nd Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	1/2	0	0	0	0

Expressing in terms of the general Pade scheme,

$$\left(\frac{\partial f}{\partial x}\right)_i = \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

4th Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	4/3	-1/3	0	0	0

Expressing in terms of the general Pade scheme,

$$\left(\frac{\partial f}{\partial x}\right)_i = 2\frac{f_{i+1} - f_{i-1}}{3\Delta x} - \frac{f_{i+2} - f_{i-2}}{12\Delta x}$$

3.0.2 Tridiagonal Schemes - Pade

6th Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	14/9	1/9	0	1/3	0

Expressing in therms of the general Pade scheme,

$$\frac{1}{3} \left(\frac{\partial f}{\partial x} \right)_{i+1} + \left(\frac{\partial f}{\partial x} \right)_i + \frac{1}{3} \left(\frac{\partial f}{\partial x} \right)_{i-1} = \frac{f_{i+2} - f_{i-2}}{36\Delta x} + 7 \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$

8th Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	19/12	1/6	0	3/8	0

Expressing in therms of the general Pade scheme,

$$\frac{3}{8} \left(\frac{\partial f}{\partial x} \right)_{i+1} + \left(\frac{\partial f}{\partial x} \right)_i + \frac{3}{8} \left(\frac{\partial f}{\partial x} \right)_{i-1} = \frac{f_{i+2} - f_{i-2}}{24\Delta x} + 19 \frac{f_{i+1} - f_{i-1}}{24\Delta x}$$

3.0.3 Pentadiagonal Schemes - Pade

8th Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	40/27	25/54	0	4/9	-1/36

Expressing in therms of the general Pade scheme,

$$\frac{4}{9} \left(\frac{\partial f}{\partial x} \right)_{i+1} - \frac{1}{36} \left(\frac{\partial f}{\partial x} \right)_{i+2} + \frac{\partial f}{\partial x}_i + \frac{4}{9} \left(\frac{\partial \phi}{\partial x} \right)_{i-1} - \frac{1}{36} \left(\frac{\partial f}{\partial x} \right)_{i-2} = 25 \frac{f_{i+2}-f_{i-2}}{216\Delta x} + 20 \frac{f_{i+1}-f_{i-1}}{27\Delta x}$$

10th Order Scheme

After comparing the coefficients on L.H.S and R.H.S using Taylor expansion, we can obtain the following.

Coefficients	a	b	c	α	β
Values	17/12	101/150	1/100	1/2	1/20

Expressing in therms of the general Pade scheme,

$$\frac{1}{20} \left(\frac{\partial f}{\partial x} \right)_{i+2} + \frac{1}{2} \left(\frac{\partial f}{\partial x} \right)_{i+1} + \frac{\partial f}{\partial x}_i + \frac{1}{2} \left(\frac{\partial \phi}{\partial x} \right)_{i-1} + \frac{1}{20} \left(\frac{\partial f}{\partial x} \right)_{i-2} = \frac{f_{i+3}-f_{i-3}}{600\Delta x} + 101 \frac{f_{i+2}-f_{i-2}}{600\Delta x} + 17 \frac{f_{i+1}-f_{i-1}}{24\Delta x}$$

3.0.4 Spectral Schemes

To create a 4th order spectral scheme, we use some differential equation constraints on the generalized Pade scheme wavenumber

$$w' = w_1, w' = w_2, w' = w_3$$

After solving the equations, we can obtain the coefficients (no optimization method was used)

Coefficients	a	b	c	α	β
Values	1.30251	0.99355	0.0375	0.577143	0.08964

Chapter 4

Graphical Comparisons between Schemes

Using MATLAB, we can compare the schemes for a series of node points. The number of nodes used is $N = 100$.

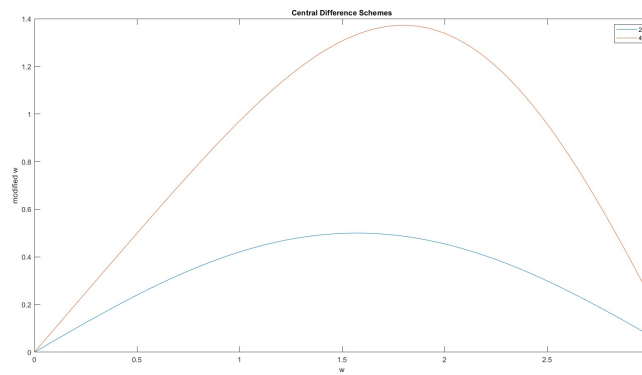


Figure 4.1: CDS w_m vs w

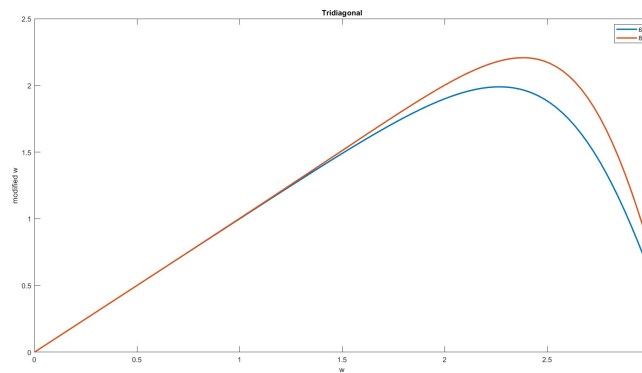


Figure 4.2: Tridiagonal schemes w_m vs w

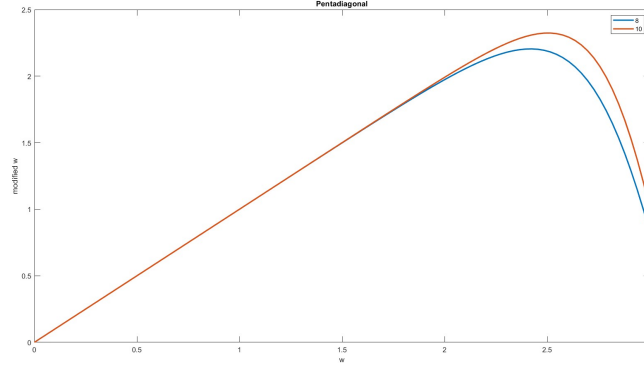


Figure 4.3: Pentadiagonal schemes w_m vs w

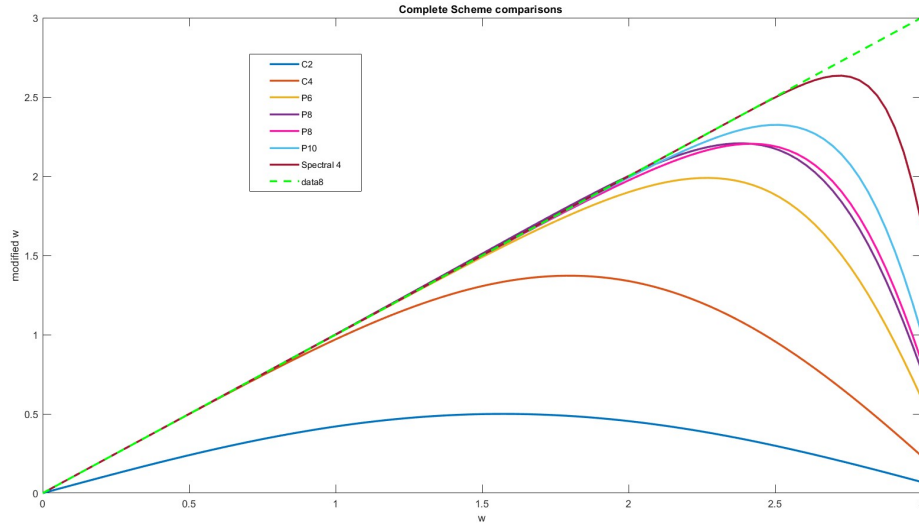


Figure 4.4: Complete Comparisons with Spectral w_m vs w

From this comparison, we can clearly see the following inferences

1. The Spectral scheme will be the closest to the actual value.
2. The compact approximations are clearly more accurate than the central difference schemes.
3. Pentadiagonal schemes are more accurate than Tridiagonal schemes of the same order.

Chapter 5

Results and Conclusions

5.1 Scheme deviations

From the final modified wavelength scheme data, we can observe the various deviations between schemes in the same order. This will help with selecting schemes since we cannot apply higher-order schemes or spectral schemes for every single application due to higher computational costs and difficulty.

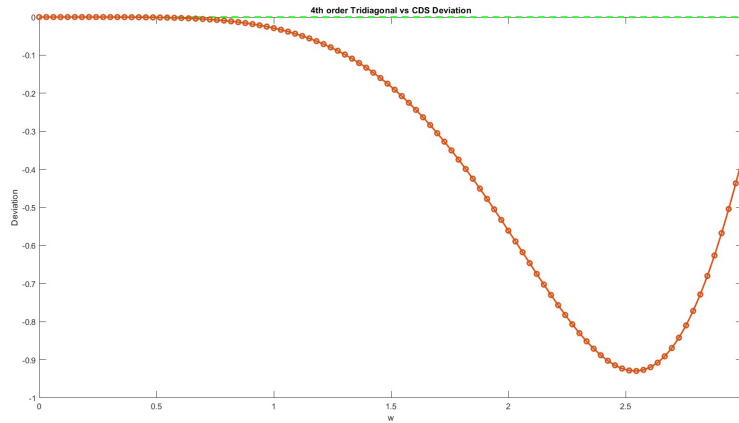


Figure 5.1: Deviation 4th order: Tridiagonal vs CDS

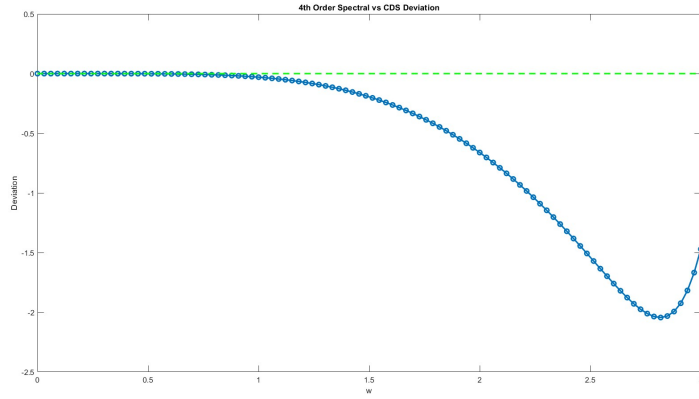


Figure 5.2: Deviation 4th order: Spectral vs CDS

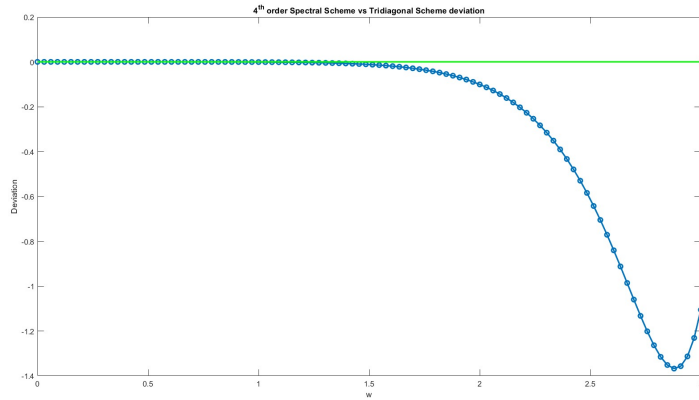


Figure 5.3: Deviation 4th order: Spectral vs Tridiagonal

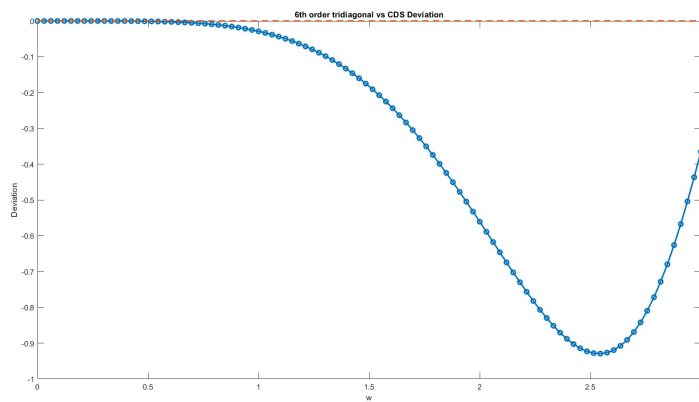


Figure 5.4: Deviation 6th order: Tridiagonal vs CDS

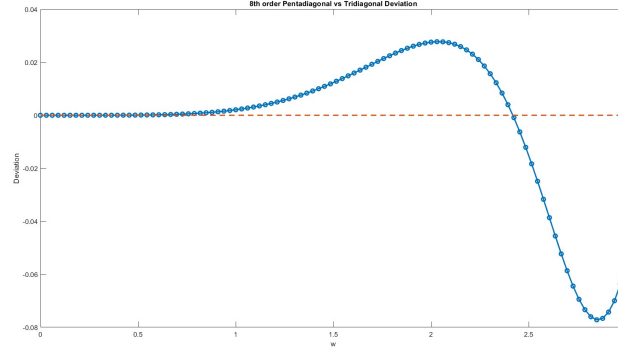


Figure 5.5: Deviation 8^{th} order: Pentadiagonal vs Tridiagonal

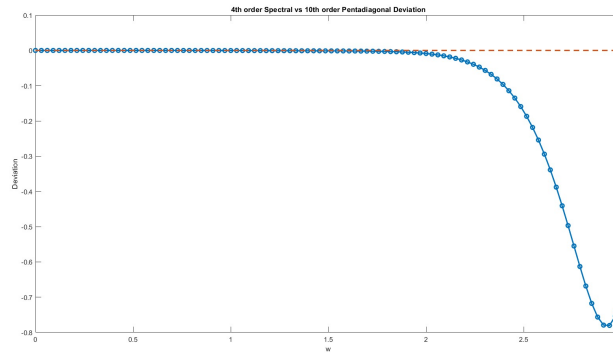


Figure 5.6: Deviation: 4^{th} Spectral vs 10^{th} Pentadiagonal

As we can see from the wavelength deviation plot for the first derivative, the central schemes are vastly inferior to the compact and spectral schemes.

When we compare the 8^{th} order Tridiagonal and Pentadiagonal schemes, we can notice that they have very little deviation. Based on the tolerances and errors allowed for the particular computation as well as the computing costs, we can decide between them.

4^{th} order Spectral vs 4^{th} order CDS showed the most deviation which is evident considering the simple approximation offered by CDS. Similarly, CDS schemes show more error even at 6^{th} order w.r.t Tridiagonal schemes. Therefore CDS schemes are usually not used in accurate computations

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- [2] David Gottlieb and Steven A. Orszag *Numerical Analysis of Spectral Methods: Theory and Applications* (Capital City Press, Montpelier, Vermont, U.S.A, 1977)
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Appendix A

MATLAB Code

A.1 For CDS

```
%Finite Difference comparisons
```

```
% %2nd order
```

```
% a = 1/2;
```

```
% b =0;
```

```
% c =0;
```

```
% alp= 0;
```

```
% bet =0;
```

```
%4th order CDS
```

```
a = 4/3;
```

```
b =-1/3;
```

```
c =0;
```

```
alp= 0;
```

```
bet =0;
```

```
%Wave number comparisons
```

```
w = linspace(0,3,100);
```

```
% w_m = (a*sin(w) + (b/2)*sin(2*w) + (c/3)*sin(3*w))/(1+2*alp*cos(w) +
```

```
w_m = (a*sin(w)+(b/2)*sin(2*w)+(c/3)*sin(3*w))./(1+ 2*alp*cos(w)+ 2*be
```

```
plot(w, w_m);
```

```
hold on;
```

A.2 For Tridiagonal schemes

```
%Modified wavenumber – Fourier error
```

```
%Transfer function
```

```
%Tridiagonal Schemes (bet = 0)
```

```
%4th order tridiagonal schemes
```

```
% %6th order tridiagonal scheme
```

```
% a = 14/9;
```

```
% b =1/9;
```

```
% c =0;
```

```
% alp= 1/3;
```

```
% bet =0;
```

```
%8th order tridiagonal scheme
```

```
a = 19/12;
```

```
b =1/6;
```

```
c =0;
```

```
alp= 3/8;
```

```
bet =0;
```

```
%Wave number comparisons
```

```
w = linspace(0,3,100);
```

```
% w_m = (a*sin(w) + (b/2)*sin(2*w) + (c/3)*sin(3*w))/(1+2*alp*cos(w) + 2*bet*cos(w));
```

```
w_m = (a*sin(w)+(b/2)*sin(2*w)+(c/3)*sin(3*w))./(1+ 2*alp*cos(w)+ 2*bet*cos(w));
```

```
plot(w, w_m);
```



```
hold on;
```

A.3 For Pentadiagonal schemes and 4th order Spectral

```
% %8th order pentadiagonal scheme
```

```
% a = 40/27;
```

```
% b =25/54;
```

```
% c =0;
```

```
% alp= 4/9;
```

```
% bet =1/36;
```

```
%10th order pentadiagonal scheme
```

```
a = 17/12;
```

```
b =101/150;
```

```
c =1/100;
```

```
alp= 1/2;
```

```
bet =1/20;
```

```
%Spectral 4th order
```

```
a = 1.30251;
```

```
b =0.99355;
```

```
c = 0.03750;
```

```
alp= 0.577143;
```

```
bet =0.08964;
```

```
%Wave number comparisons
```

```
w = linspace(0,3,100);
```

```
% w_m = (a*sin(w) + (b/2)*sin(2*w) + (c/3)*sin(3*w))/(1+2*alp*cos(w) +
```

```
w_m = (a*sin(w)+(b/2)*sin(2*w)+(c/3)*sin(3*w))./(1+ 2*alp*cos(w)+ 2*be
```

```
plot(w, w_m);
```

```
hold on;
```

